Exploring the Possibilities of Explainable AI on Low-Power Edge Devices

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Introduction to the Problem Area

- **Problem Domain: The Edge-XAI Gap**
  - AI surveillance relies on real-time, low-latency object detection on resource-constrained **Edge Devices** (e.g., Raspberry Pi).
  - Traditional models are "black boxes," lacking the **transparency** and **trust** required for high-stakes security decisions.
- **Importance in Real-World Applications**
  - **Trust & Accountability:** XAI provides an auditable trail for critical events (e.g., alarms), essential for debugging and legal compliance.
  - **Performance:** Edge processing ensures ultra-low latency and bandwidth efficiency, vital for real-time security response.
- **Motivation: Bridging the Trade-Off**
  - High-fidelity XAI methods (like LIME) are too **computationally expensive** for real-time edge deployment.
  - We aim to create a **Trustworthy, Practical Solution** by using **Uncertainty-Triggered, Selective XAI** that only runs when human review is most critical.

Problem Context: The Current Limitations

- **Existing Systems: Black-Box Object Detection at the Edge**
  - Modern surveillance relies on fast, compact models (e.g., MobileNet) running on resource-constrained hardware (e.g., Raspberry Pi).
  - These systems prioritize **speed** and **throughput** to maintain real-time performance.
- **Core Limitations and Inefficiencies**
  - **Lack of Transparency:** Models act as "black boxes," providing predictions without justifying their reasoning.
  - **Computational Conflict:** Post-hoc XAI methods (like LIME) are **computationally prohibitive** to run continuously or in real-time on edge CPUs.
  - **Resource Waste:** Running full XAI on every frame is inefficient and interrupts the critical object detection loop.
- **Challenges Faced by Users and Stakeholders**
  - **Operator Distrust:** Security personnel lack confidence when the model is uncertain, leading to potential alert fatigue or missed critical events.
  - **Debugging Difficulty:** When the model fails (e.g., misclassification under poor lighting), there is no mechanism to quickly diagnose the failure mode.

Motivation: Why Selective XAI is Necessary

- **Why This Problem is Worth Addressing (Ethical and Safety Imperative)**
  - **Trust in AI Decisions:** In high-stakes environments (security, safety), an unaccountable "black box" is a liability. Explanations transform AI into a **trustworthy decision-support tool**.
  - **Reducing False Positives:** Visual explanations allow human operators to quickly validate or dismiss uncertain alerts, minimizing alert fatigue and ensuring focus on true threats.
- **Gaps in Existing Solutions (The Edge-XAI Disconnect)**
  - **Unconstrained XAI is Non-Viable:** Running traditional LIME/SHAP continuously on an edge device (like a Raspberry Pi) destroys real-time performance due to high computational overhead.
  - **Trade-Off Failure:** Current solutions force a choice between high-speed but opaque systems, or slow but transparent systems. There is no existing method that achieves both.
- **Expected Benefits of the Proposed Solution**
  - **Resource Optimization:** By selectively triggering LIME only when model confidence falls within a defined **Uncertainty Margin**, computational resources are conserved.

Problem Statement

Core Objective

To develop a real-time edge-based surveillance camera that uses lightweight object detection and selectively triggers LIME-based explainable AI explanations during uncertain predictions to enhance transparency and trust.

Proposed Solution Overview

- **1. High-Level Idea of the Proposed System**
  - A smart surveillance camera system running on a low-power edge device (Raspberry Pi).
  - Uses an AI model to detect objects in real time from a live camera feed.
  - Adds Explainable AI (XAI) to help users understand uncertain AI decisions.
  - Explanations are generated only when the AI is unsure to save computation.
- **2. Key Features and Functionalities**
  - Real-time object detection using a lightweight AI model.
  - Confidence-based trigger to decide when explanations are required.
  - Explainable AI visualizations (e.g., heatmaps) for uncertain detections.
  - Asynchronous processing to ensure uninterrupted video streaming.
- **3. Target Users of the System**
  - Security personnel monitoring surveillance camera feeds.
  - Researchers and students working on Explainable AI and Edge AI.
  - Organizations interested in trustworthy and interpretable surveillance solutions.

System Requirements

1. **Software:**
   - Programming Language: Python
   - Frameworks: TensorFlow Lite, Flask
   - Operating System: Raspberry Pi OS (Linux)
2. **Hardware:**
   - Processor: Raspberry Pi 4 / Raspberry Pi 5
   - Memory: 4 GB RAM
   - Storage: 16 GB SD Card
   - Camera: Raspberry Pi Camera Module or USB Camera
3. **Tools:**
   - IDE: VS Code
   - Libraries: OpenCV, NumPy, LIME
4. **Deployment Platform:**
   - Web application hosted on Raspberry Pi
   - Accessed via browser over local network

Conclusion

- **1. Summary of the Problem and Proposed Solution**
  - Users cannot understand why object detections are made, especially during uncertain predictions.
  - Explainable AI methods such as LIME are computationally expensive for low-power devices.
  - Explanations are triggered only for uncertain detections and generated asynchronously, ensuring transparency without affecting real-time performance.
- **2. Feasibility of Completing the Project in the Given Timeline**
  - Uses existing lightweight object detection models and XAI libraries..
  - Modular system design enables step-by-step implementation and testing, therefore its feasible.
- **3. Scope for Future Enhancements**
  - Explore faster and more efficient XAI methods as alternatives to LIME.
  - Improve explanation speed using hardware acceleration or optimized models.
  - Integrate user feedback to further enhance trust and system reliability.

References

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Thank You